A Validated Black-Box Instrument for Language-Model Dynamics: Reproducing Known Metrics with a Token-Lattice Cellular Automaton

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Abstract

We turn a language model into a *cellular automaton over token space* — a ring of token cells resampled in place by the model's own windowed conditional $p_r(x_i \mid x_{i\pm r})$, probed by common-random-number (CRN) damage spreading — and develop it into a *validated* measurement instrument. The organizing principle is **validation by reproduction**. On the Domany–Kinzel stochastic automaton, where the damage field is provably the automaton itself, an independent prediction matches ours **bit-exactly** — zero mismatching cells over 4096×1500 updates, with a nonzero off-line control — calibrating the apparatus with no error bar attached; the instrument also separates elementary CA rules by whether damage survives (d=3.0) and recovers known Markov transition matrices. Applied across Pythia-410m training checkpoints it detects a **developmental transition**: the token-space Lyapunov exponent goes from seeds disagreeing about its *sign* to **every one of 48 post-transition runs being positive**, at two lattice sizes, with all four members of a pre-registered family surviving Benjamini–Hochberg correction. We then *delimit* the instrument, which is the result we consider most useful: real autoregressive generation absorbs *no* injected token error (P_{persist}=1.000 on three models), because free generation never resamples a token. These quantities characterise an iterated-resampling construction, not deployed decoding — a boundary we draw rather than leave to a reader.

1 Introduction

Language models are evaluated almost entirely as functions: given a context, how good is the next-token distribution? But a masked or autoregressive conditional also *defines a dynamics* — iterate it in place over a lattice of tokens and one obtains a stochastic cellular automaton whose rule is the model. Iterated-map views of LLMs have appeared qualitatively [Utimula, 2026, Perez et al., 2024, Wang et al., 2025] and as mixing-time studies of masked-LM Glauber dynamics [Sana et al., 2026]; the phenomena a CA vocabulary names — damage spreading, light cones, Lyapunov exponents, attractor censuses [Hanson and Crutchfield, 1997] — are decades old. What has been missing is not a phenomenon but a *measurement instrument whose readings can be trusted*.

Our organizing principle is therefore **validation by reproduction** (§3): before turning the apparatus on a language model we require it to recover values that are independently known. This is not a formality — building the ladder forced us to retract several of our own earlier claims, and the strongest rung is one where agreement is *exact* rather than fitted. Damage-spreading implementations are conventionally validated against fitted critical exponents, which a wrong implementation can match for the wrong reasons; a bit-exact identity cannot be passed by a plausible-looking estimator. We use only *simple, mature* primitives [Bagnoli et al., 1992, Lieb and Robinson, 1972, Hanson and Crutchfield, 1997] rather than bespoke machinery, precisely because their behaviour on known systems is a fixed benchmark. Only then do we report what the instrument measures on trained LMs (§4) and, equally, where its readings provably stop applying (§5, §6). The criticality ideas the results touch are old [Langton, 1990, Bertschinger and Natschläger, 2004, Zhang et al., 2025]; our contribution is a calibrated black-box *measurement* of them and an explicit statement of its limits.

2 The instrument

Rule. A window of $w{=}2r{+}1$ tokens with the centre masked is fed to a masked LM; the tempered centre distribution is the kernel $p_r(x_i \mid x_{i\pm r})$. For an autoregressive model the window is the r cells to the left. **Automaton.** N ring cells, async random-order single-site Glauber; special tokens forbidden as emissions. **CRN coupling.** Twin lattices sharing initial state, update order and the uniform stream differ only in the factor under test, so the *null* arm must diverge by exactly zero — asserted on every backend in the test suite, because every damage number depends on it. **Confounds.** A lattice looping one corpus n-gram scores overlap for free, so structure uses a repetition-robust variant; a degenerate low-entropy lattice snaps a perturbation back trivially, so damage D is normalised by the model's own diversity floor D_0 (twins under *independent* noise, no flip), giving $D_{\mathrm{norm}}{=}D/D_0$. Numerator and denominator necessarily use different couplings — a floor sharing the numerator's coupling is identically zero, because identical twins stay identical — but the cost is bounded: sweeping the fraction of shared draws from 0 to 0.9 moves D_{norm} by 4%. A free control disciplines our language: the temperature "phase transition" is, on a finite-size scan, a *crossover* ($\chi_{\mathrm{peak}} \propto 1/N$).

3 Validation: reproducing known metrics

We check the apparatus against systems where the answer is known, and are explicit about which properties each rung does and does not exercise (Fig. 1). The ladder is ordered by how much of the instrument's regime the rung shares: smooth-limit arithmetic first, then *discrete* state with a finite perturbation—the regime the token instrument occupies.

(1–2) Smooth-limit unit tests, labelled as such. A renormalized tangent estimator recovers the logistic map's analytic exponent to machine precision, and a coupled-map lattice matches an exact Benettin reference to 1.1×10^{-3}. Neither is evidence about damage spreading — the twin is re-anchored each step, so the estimator *is* a finite-difference derivative of what it is compared against, and **a token flip is** $O(1)$.

(3) Elementary CA: a coarse split, decisively. Driving the identical protocol with 19 rules of *known* class (12 seeds, rule as the unit of analysis) separates rules whose damage *survives* from those whose damage *dies*. The discriminating quantity is the **ignition probability**, not λ: damage in a discrete CA is bimodal, so a λ averaged over ignited and extinguished runs measures the mixture. It is also the directed-percolation order parameter [Grassberger, 1995]. Ordered rules sit at 0.046 (CI $[0.000, 0.102]$) against 0.668 (edge) and 0.682 (chaotic): **ordered vs. rest** $p{=}0.0$, $d{=}3.03$. The three-class ordering does *not* survive ($p{=}0.470$), so we claim only the coarse split — and we report this rather than the group-mean λ ordering we first computed, whose ordered end is not a measurement: five of seven ordered rules return exactly $-0.4 \ln 10$, the constant a log-linear fit yields when damage dies immediately.

(4) Domany–Kinzel: an *exact* check on the damage machinery. The DK automaton [Domany and Kinzel, 1984] is the only rung that is stochastic *and* discrete—the instrument's own regime. On its $p_2{=}0$ line the damage field between two commonly-driven replicas is provably *itself* a DK automaton at the same p_1 [Kohring and Schreckenberg, 1992]: with $s'_i = (s_{i-1} \oplus s_{i+1})\, \theta(p_1 - z_i)$, sharing z_i gives $d'_i = (d_{i-1} \oplus d_{i+1})\, \theta(p_1 - z_i)$. We therefore predict the damage trajectory with an independent simulator and demand *bit-identity*: across $p_1 \in \{0.2, \ldots, 1.0\}$ on a ring of 4096 over 1500 steps, **zero mismatching cells**, with 16 in an off-line control at $(0.6, 0.5)$ where the mapping must break. This runs through the same simulation loop that produces every language-model number below, so the window indexing, the shared-uniform consumption order, the sampling and the update schedule are all verified with no error bar attached. A second closed form rides inside the same run: at $p_1{=}1$ the update is deterministic $\oplus$ (Rule 90), so the damage field must hold exactly $2^{\mathrm{popcount}(t)}$

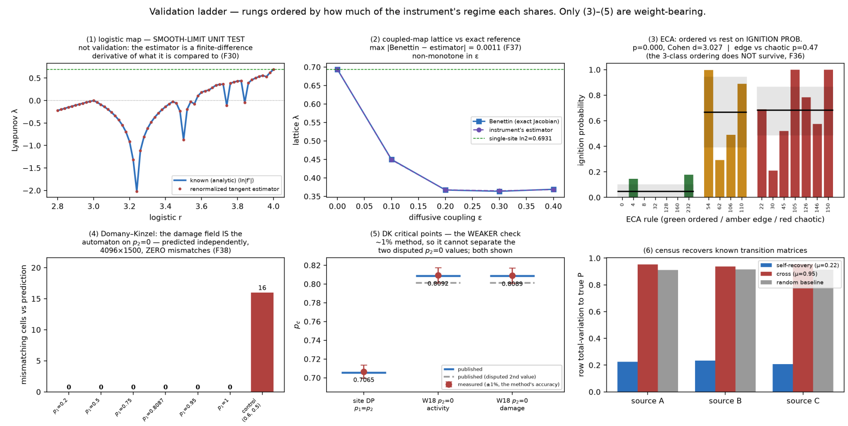

Figure 1: Validation ladder, ordered by how much of the instrument's regime each rung shares; only (3)–(5) are weight-bearing. Panel (4) is the load-bearing one: the DK damage field is predicted by an independent simulator and matches cell-for-cell, with the off-line control beside it showing the test is not vacuous.

live cells—at t=1500, $128/4096 = 0.03125$, which is what we obtain. Critical points are a weaker, $\sim 1\%$ check: p_c=0.7065 on the site-DP line (published 0.705489(4)) and 0.8092 on p_2=0, with the damage field measured *independently* at 0.8089 — coinciding with the activity transition to 3×10^{-4}, as the mapping requires. The published p_2=0 value is itself disputed (0.801(2) [Zebende and Penna, 1994] vs. 0.8087(5) [Hinrichsen et al., 1997]) and our accuracy cannot separate them; we report both and claim neither.

(5) Synthetic Markov sources. The attractor census recovers each model's own sparse transition matrix (row total-variation 0.22) and not the others (cross 0.95; random baseline 0.91). Rungs (3)–(5) share the instrument's regime and license its use on language models; (4) is the only one where agreement is exact rather than fitted.

4 A developmental transition in token-space criticality

Applied across Pythia-410m checkpoints (`step256`–`step143000`, 8 seeds, lattice sizes N=48 and 96; Fig. 2), the calibrated instrument finds the model's token-space dynamics changing from sub-critical to super-critical between steps 512 and 2000, and staying there for the rest of training.

The effect occupies a temperature window, and the window has a mechanism. Repeating the two ends of the curve at $T \in \{0.3, 0.5, 0.9, 1.1\}$ (8 seeds each, BH-FDR over the four) recovers the transition at T=0.5 (p_{BH}=$6{\times}10^{-4}$) alongside the T=0.7 of the main grid, and at none of 0.3, 0.9, 1.1 (p_{BH}=0.59, 0.72, 0.59). The ignition fraction makes that a floor and a ceiling rather than a failure: at T=0.3 damage barely propagates at either end (0.20 → 0.21) and λ_{ca} hugs zero, whereas at T=0.9 and 1.1 the lattice is *already* super-critical before the training being measured (λ_{ca} +0.19 and +0.30 at the pre checkpoint, ignition 0.65 and 0.98), so nothing is left to move. Inside the window that fraction swings hard, 0.23 → 0.81 at T=0.5. Independent work puts the critical temperature of Pythia *generation* at T_c≈1 [Nakaishi et al., 2024]; that parameterises a different sampler from our in-place update, so we take it as a consistent reference point rather than the same measurement — but a ceiling setting in between 0.7 and 0.9 is where it would predict one.

We pre-registered the test (post- vs pre-transition checkpoints, run-level Mann–Whitney, family $= \{\lambda_{\mathrm{ca}}, D_{\mathrm{norm}}\} \times \{N{=}48, 96\}$, BH-FDR) and a demotion rule: failure at either size demotes the headline. **All four members survive** ($p_{\mathrm{BH}} \leq 2 \times 10^{-5}$). From the pre-registered pre set (256/512) to the settled plateau (2000/8000/143000), λ_{ca} moves +0.0247 → +0.1683 at N=48 (d=1.59) and +0.0320 → +0.1686 at N=96 (d=1.71, n_{pre}=15: one run is excluded because its damage never ignited, so λ_{ca} is undefined for it).

The claim is better stated without any pre/post split, since every effect size depends on where that line is drawn. Before the transition the seeds do not agree on the *sign* of λ_{ca} (6/16 negative

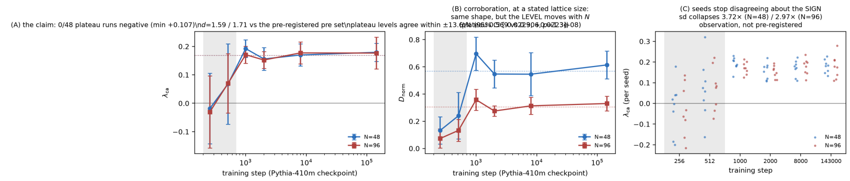

Figure 2: The developmental transition. (A) λ_{ca} rises and settles positive, the two lattice sizes on top of each other (levels agree within $\pm 14\%$). (B) D_{norm}: same shape, size-dependent level — why it corroborates rather than carries. (C) Per-seed values: sign disagreement before, consensus after.

at N=48, 7/15 at N=96, spanning -0.216 to $+0.320$); after it, **not one of 48 plateau runs is negative**, the minimum being $+0.107$. That uses every run, is ordinal, and survived a later change to how unignited runs are handled that moved every mean in the analysis.

The curve is not a step, and both ends can be inflated. It overshoots at step 1000 by 1.4–22.4% and settles; taking the peak as the post value, or the lowest checkpoint alone as the pre value, inflates the effect (1.7× on λ_{ca} for the latter), so we use the pre-registered sets at both ends. The overshoot survives correction in one of four cells, and that cell is D_{norm} — on λ_{ca} it is $+1.4\%$ at N=96 (p_{BH}=0.78) — so non-monotonicity is largely a D_{norm} phenomenon and step 1000 is not a distinct phase.

It is not specific to the model we picked. Repeating the whole grid on four Pythia sizes (70m/160m/410m/1b × 6 checkpoints × 8 seeds, 192 runs) reproduces the transition in **all four** ($p_{\mathrm{BH}} = 0.015, 0.003, 2 \times 10^{-6}, 1.5 \times 10^{-5}$). Its *timing* moves later with size and then saturates: 70m is already super-critical at the earliest checkpoint we probe, 160m crosses between steps 128 and 256, and 410m and 1b both cross between 256 and 512. Learning rate is confounded with size across the suite (1.0×10^{-3}, 6.0×10^{-4}, 3.0×10^{-4}, 3.0×10^{-4} [Biderman et al., 2023]), and the two sizes sharing a learning rate are the two sharing a crossing bracket, so we report the timing shift as an ordering and not as a size effect. An orthogonal instrument agrees on where the onset falls: Nakaishi et al. [2024] locate the emergence of critical structure in Pythia-160m during training by power spectra and part-of-speech correlation — no damage spreading involved — at $k_c{\approx}10^2$ steps, adjacent to our 128–256 bracket for the same model. Both checkpoint grids are coarse near the onset, so we read this as agreement to within a factor of about two, not as a reproduction. The plateau *level*, by contrast, does not order with size (0.162/0.164/0.174/0.166) — with four sizes the smallest attainable permutation p is $1/4! = 0.042$, so no capacity axis could have been established here, and none is visible. We report that null because it is the closest thing in this data to a previously retracted capacity claim.

Which quantity carries the claim. λ_{ca} is size-robust and D_{norm} is not, and a third lattice size settles why. Over N=48/96/192 the λ_{ca} plateau is 0.168/0.169/0.160 — a log–log slope of $N^{-0.04}$, i.e. **intensive across a** $4\times$ **range**, varying by 5% of its mean — while D_{norm} is 0.569/0.306/0.139, a slope of $N^{-1.02}$, i.e. $1/N$. That is what the construction predicts: λ_{ca} is a cone-growth *rate* fitted before saturation, so it is intensive; D_{norm} is a density ratio whose numerator stays localised in a cone while its denominator is delocalised over every site. (At two sizes we could only state a bound: the plateau levels differed by -0.0003, 95% CI $[-0.023, +0.022]$. We give the bound rather than a non-significant p-value, since a null result is not evidence of equality.) So λ_{ca} carries the claim and D_{norm} corroborates at a stated lattice size, never as a lattice-free property. Seed-to-seed variance also collapses $\sim$3× across the transition (Levene $p \leq 1.7 \times 10^{-4}$ at both sizes); not pre-registered, reported as an observation, and note it is the checkpoint means and the two lattice sizes that agree to a few percent — the plateau per-seed CV is 22–25%.

5 What the damping length measures: free generation absorbs nothing

The instrument's central assumption — that a number measured on the ring says something about the model — we tested rather than assumed, by running the identical damage protocol on *real autoregressive generation*: two CRN twins from a prompt sharing one uniform stream, one forced to a different token at one position, both continuing freely. The null arm is asserted to diverge by exactly zero, and does.

The result is unambiguous and negative. On `pythia-70m`, `-160m` and `-410m` (32 trials each) the injected error **persists in every single trial** (P_{persist}=1.000). Token identity is harsh once an injection shifts alignment, so we also measured distributionally: the total-variation distance between the twins' next-token distributions, normalised by the TV between two *independent* continuations of the same prompt, settles at $\mathrm{TV}_{\mathrm{norm}} \approx 0.97$ — as far apart as unrelated continuations.

The mechanism is structural. **Free generation never resamples a token**: once emitted, an error stays in context permanently, whereas the ring CA revisits every site, and in-place resampling is what creates the possibility of healing. The damping length, D_{norm} and the light cone therefore characterise *the iterated-resampling construction*, and we do not offer them as predictions about decoding. This also explains a light-cone velocity that depends on r but not the model, the model-invariance of $\lambda_{\mathrm{ca}}(r)$, and the failure in §6. It does *not* touch §4: the construction is held *fixed* across checkpoints — same ring, radius, temperature, coupling, seeds — so a change measured across checkpoints cannot come from the apparatus. A fixed instrument reading differently on different models is what an instrument is for; what this result forbids is transporting the units to a process the instrument does not implement. The licence is therefore *comparative* — checkpoint against checkpoint under a construction held constant — and never absolute: we do not claim any model *is* critical, since that would require the ring, the radius and D_0 to be principled rather than merely fixed, and they are choices.

Coupling, stated precisely. Damage spreading is a property of *(model, coupling)*, not of the model alone — a published result in stochastic PCAs [Kohring and Schreckenberg, 1992, Grassberger, 1995]. Our inverse-CDF sampling from a shared uniform is the *monotone* coupling: on DK's binary alphabet it coincides with the maximal-correlation member of the admissible family [Hinrichsen et al., 1997], which is why the identity above is exact, but on a vocabulary it does not, so our LM damage numbers sit inside the family rather than at an extremum (measured excess disagreement 1.3–5.4%; see the supplement). What justifies the choice is replica-independence — each replica's next state depends on its own state and the shared noise alone — which a pairwise maximal coupling lacks. Relative comparisons, including §4, are unaffected: the coupling is a common mode. **One fact explains both.** Damage-spreading theory was developed on *binary* alphabets — which is why the DK rung is exact, and why its coupling result does not survive $|V| \approx 5 \times 10^4$. The strongest calibration here and the sharpest limitation have a single cause, so an imported guarantee must be checked against the alphabet.

6 The instrument's boundary: token- vs. activation-space

A black-box instrument earns its keep if its readings track something one would otherwise need internals to see. Across two model families and two within-model axes they do not: the pairing is family-dependent (Pythia r=+0.71, GPT-2 r=−0.43; the pooled p=0.025 is a Simpson's-paradox artifact), the temperature axis is confounded, and the radius axis is null because $\lambda_{\mathrm{ca}}(r)$ is model-invariant. The reason is structural — the white-box exponent is architectural (flat across training, $\approx 1/L$) and *tangent*, while λ_{ca} is *finite*. We report a clean **negative**: the two levels are distinct and non-proxying.

7 Related work

Iterated-LLM views exist qualitatively [Utimula, 2026, Perez et al., 2024, Wang et al., 2025] and rigorously for masked-LM Glauber dynamics [Sana et al., 2026]; our novelty is the measurement layer, not the framing. Edge-of-chaos measurement is canonical [Langton, 1990, Bertschinger and Natschläger, 2004] and already applied to transformers [Tomihari and Karakida, 2025] and LLMs [Zhang et al., 2025, Li et al., 2025]; we claim neither the concept nor the criticality↔capability link. Temperature-driven order–disorder transitions are studied in several prior works [Ruan et al., 2026, Sun and Haghighat, 2025, George et al., 2025, Nakaishi et al., 2024, Arnold et al., 2024]; we claim a calibrated route, not the quantity. Perturbation propagation in *activation* space [Petrova and Vejsiu, 2026, Olivieri and Pérez Rodríguez, 2026, Fernando and Guitchounts, 2026] and over compound-AI graphs [Nilayam and Nayak, 2026] is adjacent, in a different space. We avoid "repair" [McGrath et al., 2023, Rushing and Nanda, 2024] and "self-correction" [Petrova and Vejsiu, 2026], both categorically different from a spatial damping length. Tang et al. [2026] is the calibration kin; the census descends from Hanson and Crutchfield [1997].

8 Limitations

Neither construction samples the model's actual distribution — masked conditionals are globally inconsistent, and the autoregressive rule, though consistent, is resampled in place on a ring — so all claims are properties of the dynamics. Rings are small ($N \leq 384$); the developmental result rests on one model *family* (four sizes within it, and three lattice sizes, are run, so a second family is the next requirement) and on one sampling temperature, outside which the probe saturates in one direction or the other (§4). D_{norm}'s absolute scale is doubly qualified — it moves with N, and its numerator's coupling is not extremal — so only its relative behaviour is quoted. Census ground truth is synthetic only. **Statistical scope.** The developmental family is pre-registered with BH-FDR at the run level and the ECA test uses the rule as its unit; λ_{ca} statistics exclude runs whose damage never ignited, for which the quantity is undefined, with n stated. The ladder reproductions rest on no hypothesis test, and the cross-level negative on sign-consistency rather than a surviving p-value. A previously reported capacity→sensitivity axis was pseudoreplicated and is retracted.

9 Responsible use and conclusion

This instrument measures a construction built *from* a language model from sampled outputs alone. That is its practical affordance: it characterises models behind an API, where weights are unavailable and activation-space methods cannot reach, and it detects developmental phase changes across checkpoints. It creates no new generative capability. The misuse we consider most likely is a misreading of our own numbers, which is why §5 is in the body: a damping length or light cone measured here describes an in-place-resampling ring, and real generation absorbs *no* injected token error. These quantities must **not** be cited as evidence about the robustness, error tolerance or self-correction of a deployed decoder — a λ_{ca} below zero is not a safety property. All models and corpora used are publicly released and used within their licences. That boundary is the point, and it generalises: a metric should be made to reproduce known answers before it is believed about unknown ones. We offer the ladder as a method, not only as our step one — it is what demoted our own ECA ordering and caught a λ_{ca} reported where it is undefined. An instrument that does that, and marks where its readings stop applying, is more useful than one that claims more.

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A Reproducibility

Models. `bert-tiny/mini/base` (L=2–24); `Pythia-14M–1B` [Biderman et al., 2023] and `GPT-2` small–xl; Pythia-410m checkpoints `step256–step143000`. A tiny transformer trained from scratch on tinyshakespeare provides the corpus-checked census. **Grids.** $r \in \{1, 2, 4, 8, 16\}$, $T \in \{0.3, \ldots, 1.5\}$, $N \in \{48, 96, 192, 384\}$, 8 seeds for the developmental family. **Definitions.** $D_{\mathrm{norm}} = D/D_0$ with D_0 the independent-noise no-flip floor; λ_{ca} the log-slope of damaged-site count over the pre-saturation window, with the fit window's constants exposed as parameters and a named sentinel for never-ignited runs. **Tests.** The exact-zero CRN null is asserted over every backend and both update modes through one shared simulation loop; a golden-file regression asserts the core stays bit-identical; the DK identity is asserted cell-for-cell with an off-line control that must fail. All code, per-run result files, and figure scripts accompany the submission; every number in the paper is traceable to a result file.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract states the contribution as a *validated* black-box instrument and names the three things it does: reproduce known metrics before measuring LMs (the load-bearing rung being Domany–Kinzel, where agreement is *bit-exact* rather than fitted), detect a developmental transition across Pythia checkpoints, and *delimit* itself — real generation absorbs no injected token error, so the quantities describe an iterated-resampling construction rather than decoding. The smooth-limit rungs (logistic map, coupled-map lattice) are labelled in the abstract and body as arithmetic unit tests, not validation. The criticality phenomena are framed as decades-old (measured, not discovered) and cited to prior work. The cross-level proxy hypothesis is reported as a clean *negative*. No capacity scaling law is claimed — that earlier two-seed gap is retracted and explicitly not built upon.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work?

   Answer: [Yes]

   Justification: The Limitations section states: neither construction samples the model's actual distribution (masked conditionals are globally inconsistent; the autoregressive rule is consistent but resampled in place on a ring), so all claims are properties of the dynamics; small rings ($N \leq 384$) and one model family for the developmental result, with a second family named as the obvious next requirement; D_{norm}'s absolute scale doubly qualified (it moves with N, and its numerator's coupling is not extremal), so only its relative behaviour is quoted; synthetic-only census ground truth; the windowing scheme as a first-class apparatus factor; temperature scales not comparable across constructions. The statistical scope is stated separately, including that λ_{ca} statistics exclude runs whose damage never ignited — for which the quantity is undefined — with n stated in each case.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [N/A]

   Justification: The paper contains no formal theorems; it is empirical/measurement work. Connections to the Lieb–Robinson bound and the Bagnoli–Rechtman–Ruffo Lyapunov–velocity relation are cited, not re-derived.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results?

   Answer: [Yes]

   Justification: All code, model checkpoints (toy), per-run result JSON/NPZ, and figures are supplied as anonymised supplementary material; the Reproducibility appendix gives models, grids, seeds, batch/ring sizes, the diversity-floor and CRN definitions, and the statistical tests. A harness regression test asserts the CRN null coupling is exactly zero.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions?

   Answer: [Yes]

   Justification: The anonymised supplement contains the full codebase with a README giving per-phase reproduction commands and measured M1 runtime tables, pinned dependency versions, and idempotent resumable scripts. Every number in the paper is traceable to a committed result file; the analysis scripts assert their emitted group sizes against their declared pre-registered design. A public repository URL will be added at camera-ready if the paper is accepted.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details?

   Answer: [Yes]

Justification: Reproducibility appendix specifies optimizer/steps for the toy model, the exact $(r, T, N, B, \text{sweeps}, \text{seeds})$ grids per experiment, the reference corpora, and the metric definitions.

7. **Experiment statistical significance**
   Question: Does the paper report error bars / statistical significance?
   Answer: [Yes]
   Justification: The paper's central claims are the validation-ladder reproductions—exact or near-exact matches to known values (the Domany–Kinzel damage identity, which is verified *bit-exactly*—zero mismatching cells—and its critical points to 0.06–0.15%; logistic-map Lyapunov mean error $< 10^{-3}$; census self-TV 0.22 vs cross 0.95)—which are not null-hypothesis tests, so multiple comparisons do not apply to them. Seed error bars are reported where relevant. Where we *do* test—the ECA class separation and the developmental transition—the unit of analysis is the rule or the run (never the seed-cell), the tests are bootstrap or rank-based, and the developmental family is pre-registered and corrected with Benjamini–Hochberg FDR. The finer three-class ECA ordering was tested, failed (p=0.470), and is reported as not surviving. The cross-level conclusion is a *negative* reached by sign-consistency across two model families and two within-model axes, not a surviving p-value, so no multiplicity correction is load-bearing. The demoted capacity gap is disclosed as a suggestive two-seed effect and not built upon; an optional seed-level rebuild is tracked in the repository (issues #1–#2, #10).

8. **Experiments compute resources**
   Question: Does the paper provide sufficient information on compute resources?
   Answer: [Yes]
   Justification: All experiments run on a single Apple M1 Pro (16 GB); per-phase wall-clock runtimes are tabulated. The toy model uses CPU JAX; real models use fp16 on the MPS backend. Total compute is a few laptop-days.

9. **Code of ethics**
   Question: Does the research conform with the NeurIPS Code of Ethics?
   Answer: [Yes]
   Justification: Measurement/interpretability work on public pretrained models and synthetic data; no human subjects, no sensitive data, no new capability that raises misuse concerns.

10. **Broader impacts**
    Question: Does the paper discuss potential positive and negative societal impacts?
    Answer: [Yes]
    Justification: See the Responsible use section. Positive: black-box measurement of trained-LM token dynamics without weight access, applicable to closed/API-only models. It creates no new generative capability. The negative case we treat as most likely is misreading: the paper's damping length and light cone describe an in-place-resampling construction, and real autoregressive generation absorbs no injected token error, so they must not be cited as evidence about deployed-decoder robustness. That boundary is stated in the body, not an appendix.

11. **Safeguards**
    Question: Does the paper describe safeguards for responsible release of high-risk assets?
    Answer: [N/A]
    Justification: No high-risk models or datasets are released; only measurement code and analysis results on public models.

12. **Licenses for existing assets**
    Question: Are existing assets (code, data, models) properly credited and licenses respected?
    Answer: [Yes]
    Justification: All assets are publicly released and cited: Pythia and BERT checkpoints (Apache-2.0), GPT-2 (MIT), tinyshakespeare (public-domain source text), WikiText-103 (CC BY-SA 3.0), PyTorch (BSD-3-Clause), HuggingFace Transformers and JAX (Apache-2.0). Usage is research-only and within each licence.

13. **New assets**
    Question: Are new assets introduced in the paper well documented?
    Answer: [Yes]

Justification: The instrument (code), the synthetic-Markov calibration corpora, and the toy checkpoints are documented in the repository README and the Reproducibility appendix.

14. **Crowdsourcing and research with human subjects**
    Question: N/A.
    Answer: [N/A]
    Justification: No human subjects or crowdsourcing.

15. **Institutional review board (IRB) approvals**
    Answer: [N/A]
    Justification: No human-subjects research.

16. **Declaration of LLM usage**
    Question: Does the paper describe LLM usage if it was an important, original, or non-standard component?
    Answer: [Yes]
    Justification: Pretrained LLMs are the object of study — they supply the CA update rule — and this is documented throughout. Separately, and disclosed here per venue policy: an AI coding assistant was used substantially in implementing the experiments, in analysis scripting, and in drafting. All experimental designs, pre-registrations and claims were reviewed and verified by the authors against committed result files; several errors introduced during that process were caught by the repository's own assertions and are recorded as retractions in the supplementary findings log.